A. S. M Mustakim Rahman Siddique

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I build machine learning systems that learn from data they can't see all at once, and run on hardware that can't afford waste — with a research focus on federated learning under non-IID data, medical imaging, efficient foundation models, and low-resource speech.

Education

Rajshahi University of Engineering & Technology (RUET) Awarded September 05, 2023

Bachelor of Science in Computer Science & Engineering *CGPA: 3.43/4.00*

- 3.64/4.00 credit-weighted GPA over final four semesters (3rd and 4th year); First Class.
- Relevant coursework: Artificial Intelligence, Neural Networks & Fuzzy Systems, Data Mining, Digital Image Processing, Database Systems, Operating Systems, Parallel & Distributed Processing.

Research Interests

- **Federated & Collaborative Learning:** learning under non-IID and heterogeneous data, personalization, robust and privacy-preserving learning, and communication-efficient intelligence across distributed and resource-constrained systems.
- **Foundation Models & Generative AI:** adaptation, fine-tuning, reasoning, reliability, and efficient deployment of large language, multimodal, and other foundation models for domain-specific applications.
- **Efficient Machine Learning:** model compression, parameter-efficient adaptation, resource-efficient learning, and inference optimization for scalable deployment from edge devices to large-model systems.
- **Intelligent AI Systems:** scalable and reliable AI systems, model–system co-design, and integration of learning-based models into real-world applications.
- **Speech & Multimodal Intelligence:** robust and low-resource speech and language learning, multilingual and code-switched settings, multimodal representation learning, and dialect-diverse settings.

Publications

Peer-Reviewed Publications

CARBO: Clustering and Rotation-Based Oversampling for Class Imbalance Learning

- Mahit Kumar Paul, Biprodip Pal, AHM Sarowar Sattar, **A S M Mustakim Rahman Siddique**, Md Al Mehedi Hasan. *Knowledge-Based Systems*, Vol. 300, 2024, Article 112196. [Paper]

Manuscripts Under Review

FedQuaDx: Multi-Stage Quantization and Diffusion-Guided Personalization for Non-IID Collaborative Diagnosis

- **A S M Mustakim Rahman Siddique**, Biprodip Pal — *Under Review.*
- Developing a personalized federated learning framework for non-IID medical imaging data that integrates multi-stage model quantization with diffusion-guided representation learning.
- Aims to reduce communication overhead while improving client-level personalization and diagnostic accuracy in decentralized healthcare settings.

Undergraduate Thesis

Trustworthy Federated Learning via Blockchain-Based Model Aggregation and Verification 2023

- *B.Sc. Thesis, Dept. of CSE, RUET* — Supervisor: Mahit Kumar Paul
- Proposed a decentralized federated learning framework that replaces a central aggregation server with a blockchain-based consensus mechanism, removing the single point of trust/failure inherent in standard FedAvg-style architectures.
- Designed a smart-contract-based protocol for verifying and aggregating client model updates, improving auditability and resistance to malicious or corrupted client contributions.
- Evaluated the framework on MNIST, Fashion-MNIST, and CIFAR-10, analyzing trade-offs between blockchain-induced communication/latency overhead and gains in trust, transparency, and robustness compared to conventional federated learning.

Research Projects

EMCAD Replication and FedTriad: Federated Polyp Segmentation [Code]

- Reproduced EMCAD (CVPR 2024) with a PVTv2-b2 encoder on CVC-ClinicDB over five independent runs: 94.85 ± 0.37 test Dice at 50 epochs (paper: 95.21 at 200) and 10.58 G FLOPs at 352^2 (paper: 10.59 G).
- Designed FedTriad, a federated fine-tuning scheme that freezes the LayerNorm encoder, averages decoder weights, and keeps decoder BatchNorm local, cutting federated upload by 55.9× versus FedAvg.
- Found and verified a height/width transposition bug in the polyp data loader costing about 0.13 Dice, and showed about 93% of inference cost sits in the encoder, capping decoder-only efficiency gains near 7%.
- Technologies: PyTorch, EMCAD, PVTv2, Federated Learning, FedTriad

Federated Learning for Aircraft Engine Health Monitoring (NASA C-MAPSS) [Code]

- Designed a federated learning pipeline simulating 5 airline operators for aircraft engine prognostics, training a dual-head CNN-GRU model for Remaining Useful Life (RUL) regression and early fault detection under non-IID client data.
- Investigated 7 research questions covering heterogeneous sensor availability, client class imbalance, gradient-based interpretability, concept drift, validation mismatch, membership-inference risk, and robust aggregation against model poisoning.
- Federated model (sample-weighted FedAvg) outperformed isolated and centralized baselines: RMSE 14.02 (centralized: 16.17) and AUROC 0.997 on RUL/fault prediction; weighted BCE raised fault recall to 0.967 under class imbalance.
- Technologies: PyTorch, Federated Learning, FedAvg, CNN-GRU, Gradient Attribution, Robust Aggregation

FL-QAT: Quantized-Communication Benchmark for Federated Learning [Code]

- Built a benchmarking framework comparing vanilla FedProx-based FL against adapter-based FL under FP32, INT8, and INT4 quantized client-server communication, across three medical-imaging datasets (lung ultrasound, brain tumor, diabetic retinopathy) and backbones (MobileNetV2, ResNet18, ViT-B/16).
- Implemented quantization-aware training with error feedback and configurable uplink/downlink quantization to characterize communication-efficiency vs. accuracy trade-offs under Dirichlet-controlled non-IID client partitioning.
- Technologies: PyTorch, Quantization-Aware Training (QAT), FedProx, Dirichlet Partitioning, torchvision

Professional Experience

Vivasoft Limited Nov 2023 – Present

Software Engineer (ML/AI) *Dhaka, Bangladesh*

- Led two cross-functional engineering teams (5 and 8 engineers) delivering an AI job-crawler platform and a real-time, interruptible Bangla voice agent over WebRTC, with VAD-based segmentation, turn detection, and CUDA-level ASR/TTS optimization for streaming latency.
- Fine-tuned Whisper, Wav2Vec2, and NVIDIA NeMo Conformer with a custom Bengali tokenizer, improving WER and CER on noisy, low-bitrate call-center audio, then served the models on CPU with CTranslate2 and oneMKL.
- Built a Bengali call-center analytics pipeline combining ASR, speaker diarization, sentiment, and topic detection with automated reporting.
- Designed and shipped Tiger Agent, an enterprise RAG chatbot with document extraction and hybrid dense + BM25 search over a Weaviate vector database, streaming responses to the frontend over server-sent events.
- Used LoRA and QLoRA with quantized inference to cut memory and training cost while keeping ASR accuracy.
- Refactored Go and Python services for about 30% better performance.

Vivasoft Limited Aug 2023 – Oct 2023

Associate Software Engineer *Dhaka, Bangladesh*

- Consolidated 7 microservices into 2 core services, cutting redundant deployments and improving deployment efficiency by about 70%.
- Designed data models and transactional workflows for batch processing and inventory management, with an eye toward efficient data-access patterns.
- Hardened services for production with unit tests, structured error handling, and fault-tolerant integrations.
- Built Go and Python microservices, schedulers, and REST APIs; built email and Slack notification systems and integrated Kong as the API gateway.

Industry & Applied ML Projects

Tiger Agent: Enterprise RAG Chat System

- Developed an enterprise-grade retrieval-augmented generation (RAG) system with hybrid dense + BM25 search over a vector database, document ingestion, and prompt-based response generation, streamed to the frontend over server-sent events.
- Technologies: Python, Weaviate, LangChain, FastAPI, Server-sent events

Bengali Call-Center Audio Analytics Platform

- Built a Bengali speech analytics system combining ASR, sentiment analysis, topic detection, speaker diarization, and automated report generation for structured conversation understanding.
- Fine-tuned NeMo Conformer and Whisper Medium models with a custom Bangla tokenizer and optimized inference using CTranslate2 with oneMKL for efficient CPU deployment.
- Technologies: Python (FastAPI), Go (Gin), Whisper Medium, Conformer, Faster Whisper, MySQL, Redis

Real-Time Bangla Voice Agent

- Designed an interruptible, real-time Bangla voice assistant integrating VAD-based speech segmentation, turn detection, low-latency ASR, TTS, and RAG-based response generation over WebRTC.
- Optimized CUDA-level inference for the ASR and TTS stages to meet real-time, streaming latency constraints.
- Technologies: FastAPI, WebRTC, PyTorch, CUDA, NeMo Conformer, Piper TTS

Efficient Adaptation and Deployment of Open-Weight LLMs (Gemma 12B & E4B)

- Adapted the Gemma 12B open-weight LLM for domain-specific generative tasks, studying how fine-tuning strategy, instruction data design, and inference configuration jointly affect task performance and deployment efficiency.
- Applied LoRA and QLoRA with systematic tuning of training configurations and instruction data to specialize the model while minimizing adaptation cost, and evaluated quantized deployment with Gemma E4B.
- Technologies: PyTorch, Hugging Face Transformers, PEFT, LoRA, QLoRA, Quantization, ONNX

Technical Skills

Languages: Go, Python, CUDA C++
Frameworks: Gin, Echo, FastAPI
Libraries: PyTorch, Transformers, NumPy, Scikit-learn, Matplotlib
Databases: MySQL, PostgreSQL, MongoDB, Redis
Vector Databases: ChromaDB, Weaviate
DevOps & Tools: Git, Docker, RabbitMQ, Celery
Architectures: Microservices, Event-driven systems, Design patterns, Distributed Systems
AI Tools: LangChain, OpenAI APIs, Azure Cognitive Services, Parameter-efficient fine-tuning, LoRA, QLoRA
Cloud Platforms: Azure, AWS, GCP

Test Scores

- **IELTS:** Overall 6.5 — Listening: 8.0, Reading: 6.0, Writing: 6.5, Speaking: 6.0

Awards & Achievements

- Government Scholarship, Dhaka Education Board (awarded based on academic merit).
- First Place, Inter-Company AI Hackathon, Software Industry Competition, for developing an AI-based solution using machine learning and natural language processing techniques.
- Runner-up, Inter-Software-Company Table Tennis (Singles) Competition.

Academic Service & Leadership

- **Technical Team Lead:** Led two cross-functional engineering teams in the development of AI systems including speech processing and NLP-based applications.
- **Co-ordinator, Vivasoft AI Hackathon 2025:** Organized and managed technical execution, mentoring, and evaluation processes for participating teams.
- **Technical Point of Contact, Learnathon 3.0, Geeky Solutions (Bangladesh):** Served as technical liaison supporting participants and coordinating implementation guidance.
- **Mentor, Learnathon 2.0, Geeky Solutions (Bangladesh):** Delivered instruction on Python programming and core AI concepts, and provided hands-on mentoring during implementation of applied machine learning projects.

References

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